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Exercise No: 2

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COMBINING TWO TILES AND CREATING COLOUR COMPOSITES

Objective: To download Landsat satellite images of Malappuram district from USGS for years before 2000 and 2020, and process them in QGIS to generate false color composites.

Software Required:

- **QGIS (latest version installed)**
- **Internet access to download satellite data from USGS Earth Explorer**
- **HCMGIS Plugin for Google Maps**
- **Malappuram district shapefile (.shp)**

Procedure:

1. Data Download (Before 2000)

- Open USGS Earth Explorer.
- Select the study area: Malappuram district.
- Dataset: Landsat 4–5 Collection (data before 2000).
- Bands Available: B1, B2, B3, B4.
- Cloud Cover: 0–5%.
- Since Malappuram is covered by two satellite paths, download both images.

2. Import Shapefile

- Install and activate HCMGIS plugin for Google Maps reference.
- Open QGIS → Data Source Manager → Vector → Browse → Select (malappuram.shp) → Add → Close.

3. Add USGS Raster Files (Before 2000)

- Open QGIS → Data Source Manager → Raster → Browse → Select four bands (B1–B4) → Add.

4. Create Virtual Raster for Each Tile

- Go to Raster tab → Miscellaneous → Build Virtual Raster.
- Input Layer: Select the four bands.
- Check “Place each input file into a separate band”.
- Save Virtual Raster file → Close.
- Remove the original four bands to avoid duplication.

- Repeat the same steps for the second tile.

5. Combine Two Tiles (Mosaic)

- Go to Raster → Miscellaneous → Build Virtual Raster.
- Input Layers: Select both virtual rasters (Tile 1 and Tile 2).
- Save → Run → Close.

6. Clip Study Area

- Go to Raster → Extraction → Clip Raster by Mask Layer.
- Input Layer: Combined raster.
- Mask Layer: Malappuram shapefile.
- Save → Run.

7. Create False Color Composite (Before 2000)

- Right-click the clipped raster → Properties → Symbology.
- Assign: Red = Band 4, Green = Band 3, Blue = Band 2.
- Apply → OK.
- This gives a false color image where vegetation appears red.

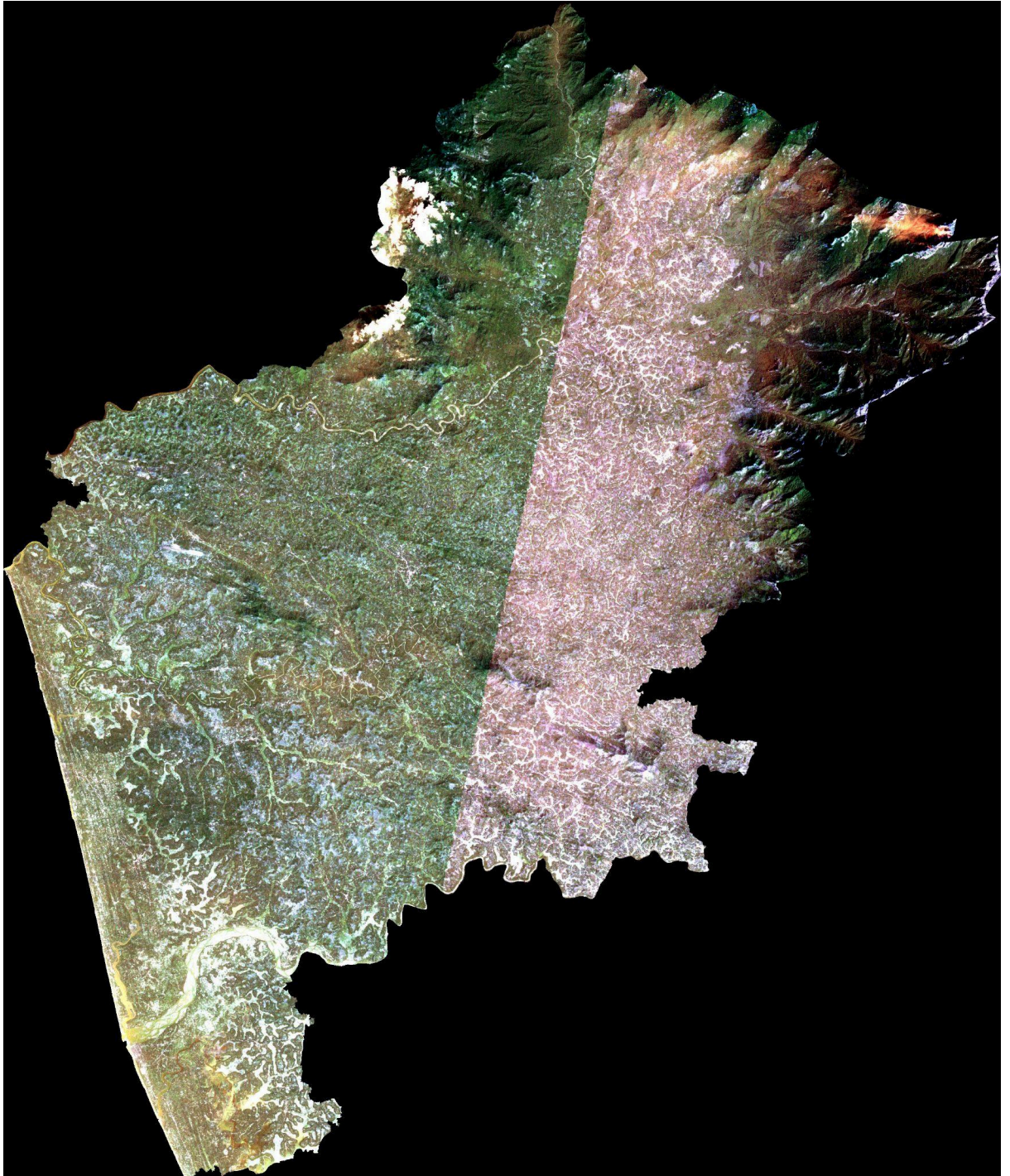
8. Repeat for the Year 2020

- Follow the same procedure using Landsat 8/9 Collection 2 Level 2 data (Bands B2, B3, B4, B5).
- Generate a false color composite (B4, B3, B2) for comparison with the pre-2000 data.

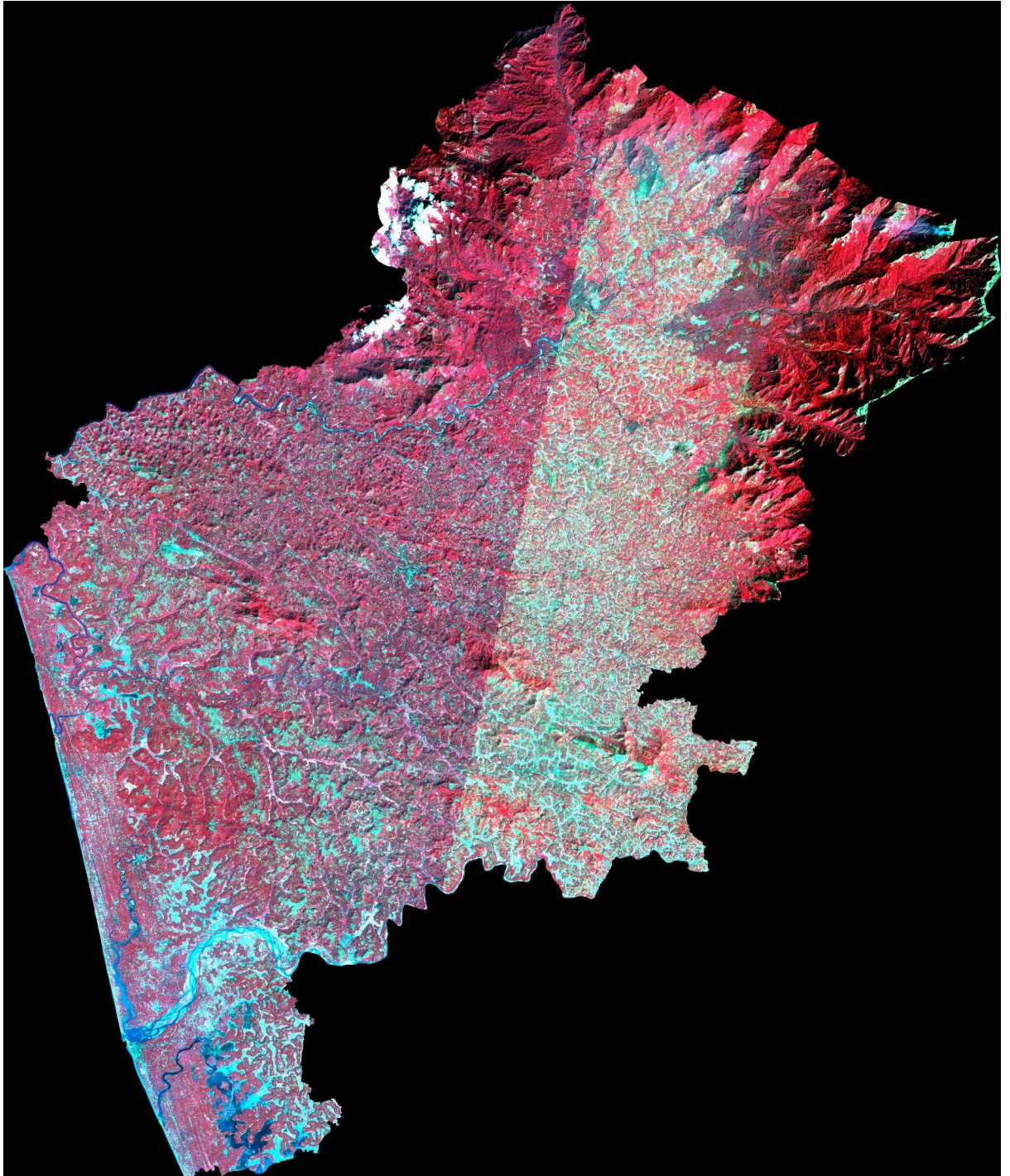
Results:

1. Generated **True Color Composite (B3, B2, B1)** for Malappuram district before 2000.

Image 1

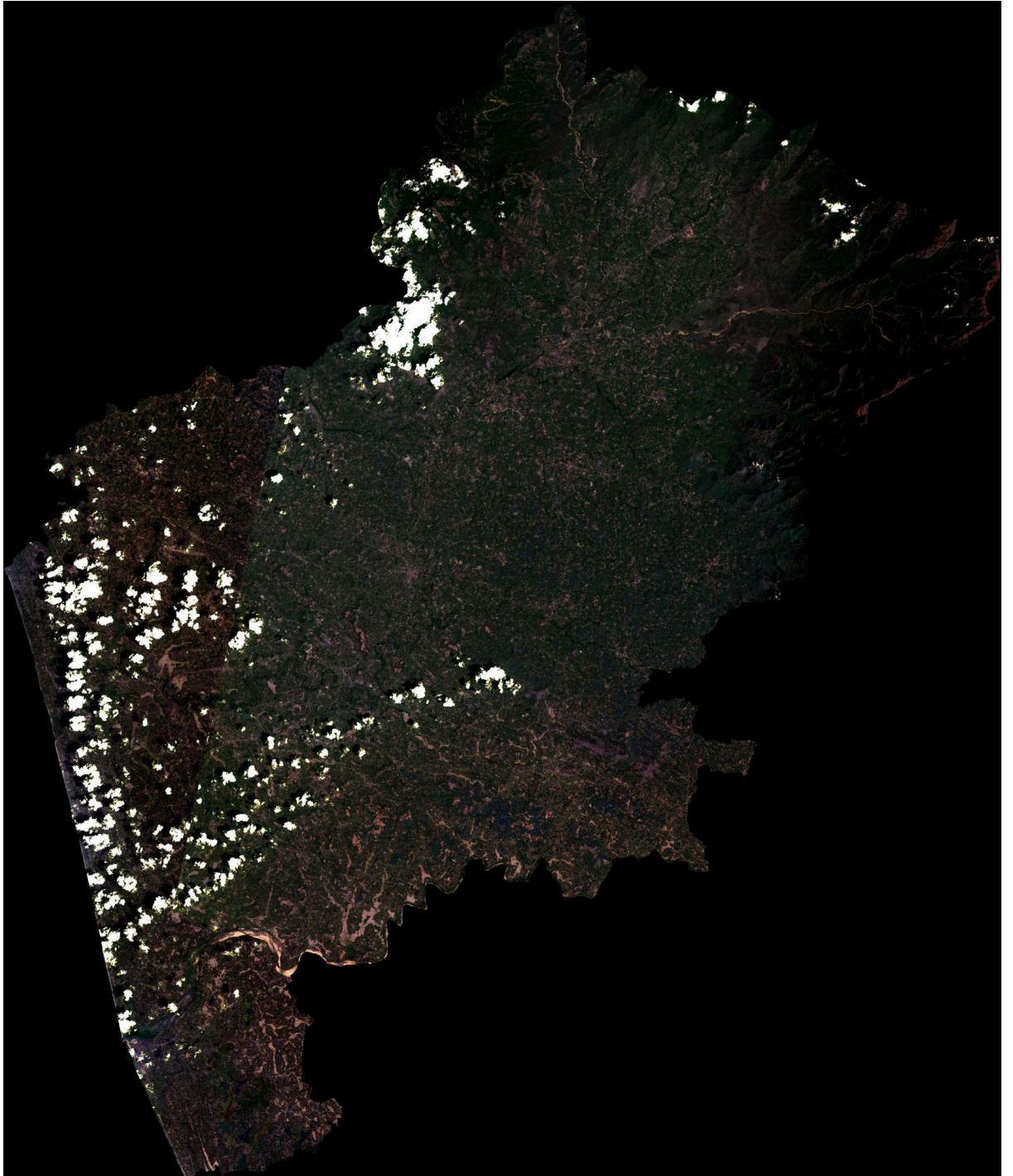


2. Generated **False Color Composite (B4, B3, B2)** for Malappuram district before 2000.
Image 2



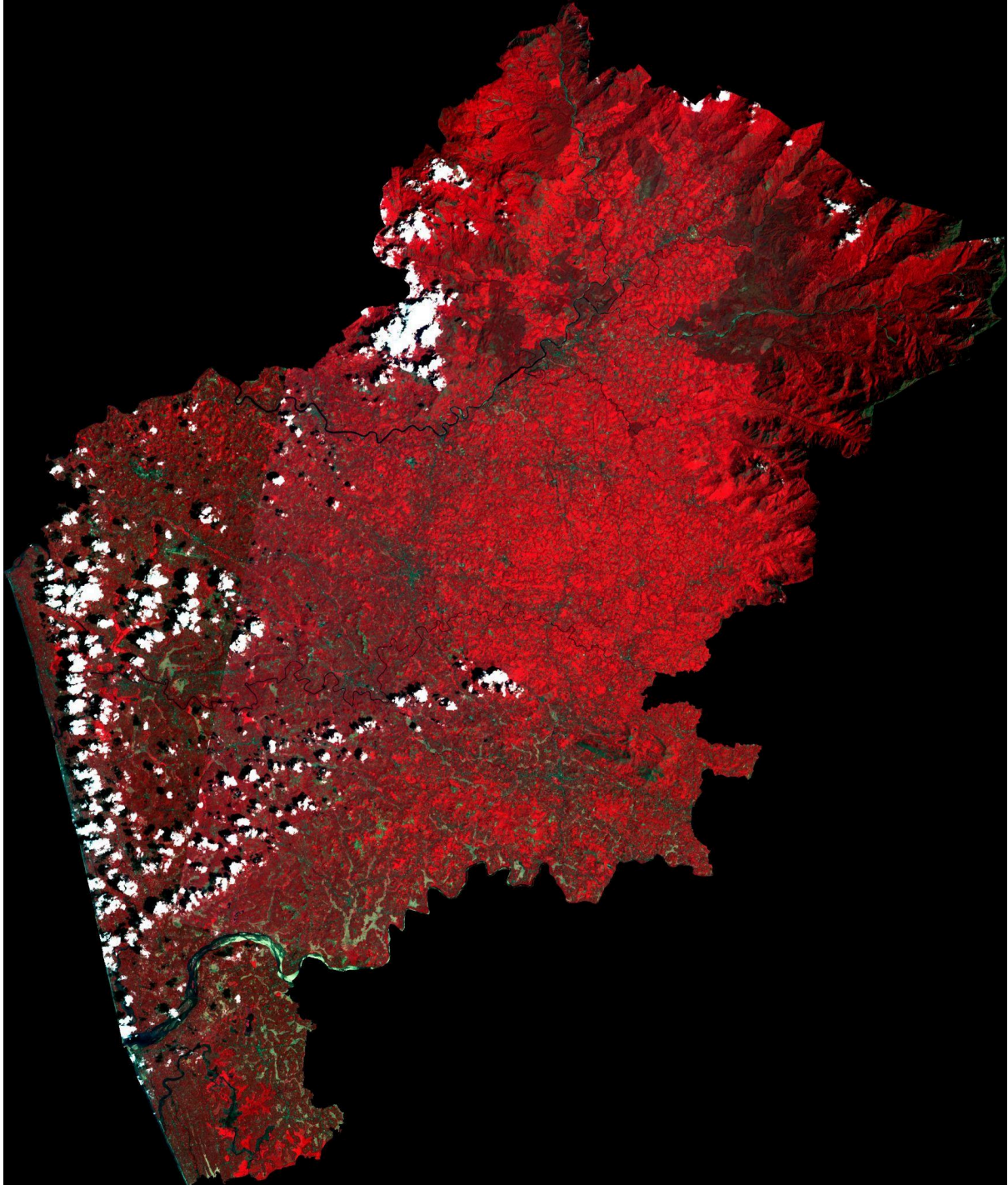
3. Generated **True Color Composite (B3, B2, B1)** for Malappuram district in 2020.

Image 3



4. Generated **False Color Composite (B4, B3, B2)** for Malappuram district in 2020.

Image 4



Conclusions:

The exercise demonstrated how to download and preprocess satellite imagery in QGIS using virtual rasters, mosaicking, and clipping.

- Data before 2000 was obtained from **Landsat 4–5 (Bands 1–4)**.
- Data for 2020 was obtained from **Landsat 8/9 (Bands 2–5)**.
- False color composites effectively highlight vegetation and land use changes across years.

Submitted by

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